

Matgeo-2.10.8

AI25BTECH11019 - MENAVATH SAI SANJANA

Question:

Let $\mathbf{b} = 4\hat{i} + 3\hat{j}$ and $\mathbf{c}$ be two vectors perpendicular to each other in the XY plane. All vectors in the same plane having projections 1 and 2 along $\mathbf{b}$ and $\mathbf{c}$, respectively, are given by .

Solution:

Consider

$$\mathbf{r} = \begin{pmatrix} r_1 \\ r_2 \end{pmatrix}, \quad (0.1)$$

where r_1 and r_2 are the x- and y-components of the required vector.

$$\mathbf{b} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix} x \\ y \end{pmatrix} \quad (\text{perpendicular to } \mathbf{b}) \quad (0.2)$$

$$\text{Scalar projection on } \mathbf{b} : \frac{\mathbf{r}^T \mathbf{b}}{\|\mathbf{b}\|} = 1 \Rightarrow \mathbf{r}^T \mathbf{b} = 5 \quad (0.3)$$

$$\text{Since } \mathbf{c} \perp \mathbf{b}, \quad 4x + 3y = 0 \Rightarrow x = -3t, y = 4t \quad (0.4)$$

$$\text{Scalar projection on } \mathbf{c} : \frac{\mathbf{r}^T \mathbf{c}}{\|\mathbf{c}\|} = 2 \Rightarrow \mathbf{r}^T \mathbf{c} = 10t \quad (0.5)$$

$$\mathbf{r}^T \mathbf{b} = 4r_1 + 3r_2 = 5 \quad (0.6)$$

$$\mathbf{r}^T \mathbf{c} = -3r_1 + 4r_2 = 10 \quad (0.7)$$

$$\text{Solving: } r_1 = -\frac{2}{5}, \quad r_2 = \frac{11}{5} \quad (0.8)$$

Conclusion: The required vector is

$$\mathbf{r} = \begin{pmatrix} -\frac{2}{5} \\ \frac{11}{5} \end{pmatrix}. \quad (0.9)$$

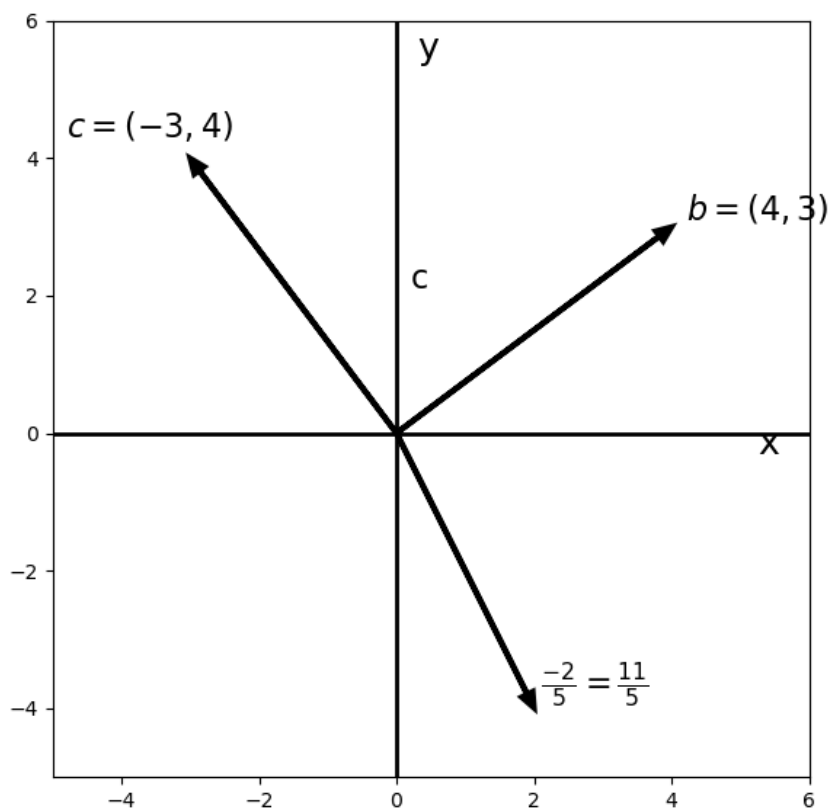


Fig. 0.1